

AI Data Center Sustainability & Energy Intelligence

Where is AI infrastructure being built, and does it line up with electricity cost, carbon intensity, and renewable availability worldwide?

THE PROBLEM

AI's compute boom has a resource cost

Training and running frontier AI models requires enormous GPU clusters, concentrated in data centers that consume:

Electricity

Huge and growing power draw — single sites now exceed 1 GW

Water

Large cooling volumes, depending on site design

Capital

Tens of billions of dollars in build-out costs

Chips

Thousands to hundreds of thousands of GPUs per site

The question this project answers:

“As AI adoption accelerates, what factors make a country an ideal location for efficient and sustainable AI data centers — and does today's build-out actually reflect those factors?”

DATA

No single dataset covers this — four sources, combined



Epoch AI — AI Data Centers

CC BY

6 datasets: data centers, GPU clusters, timelines, chip deployments, chillers, cooling towers



Our World in Data — Energy

Open, CC BY

Renewable share of electricity + carbon intensity, by country/year



Our World in Data — CO2

Open, CC BY

Total CO2 emissions + per-capita, by country/year



GlobalPetrolPrices.com

Manually compiled snapshot

Residential electricity price by country (USD/kWh)

Merging required aligning inconsistent country-name spellings across all four sources (formal ISO names vs. common names) — handled once in the cleaning pipeline via a shared join key.

Methodology

1. Clean

Raw → processed pipeline
(clean_data_centers.ipynb): coerce types, snake_case columns, split confidence tags, filter non-country aggregate rows, align country naming across all 4 sources.

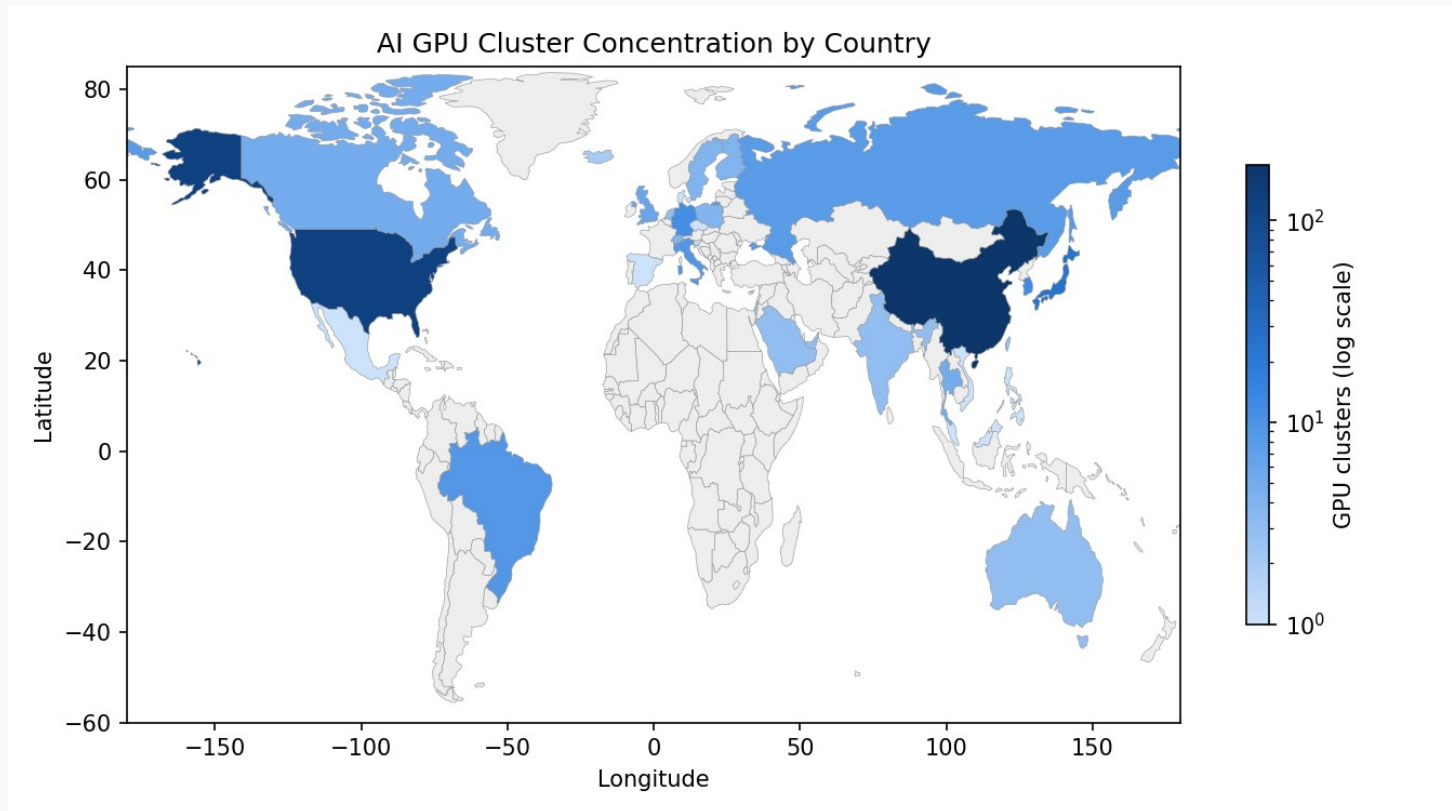
2. Analyze

10 analytical questions (02_EDA.ipynb), each with the chart type the question calls for — choropleth, regression, K-Means clustering, treemap, ranking — and a written finding.

3. Present

Streamlit dashboard (5 tabs) for interactive exploration: filter by operator/country, compare countries side-by-side, drill into any of the 10 findings.

China outnumbered the US on AI cluster count — not just power



China: 188 GPU clusters

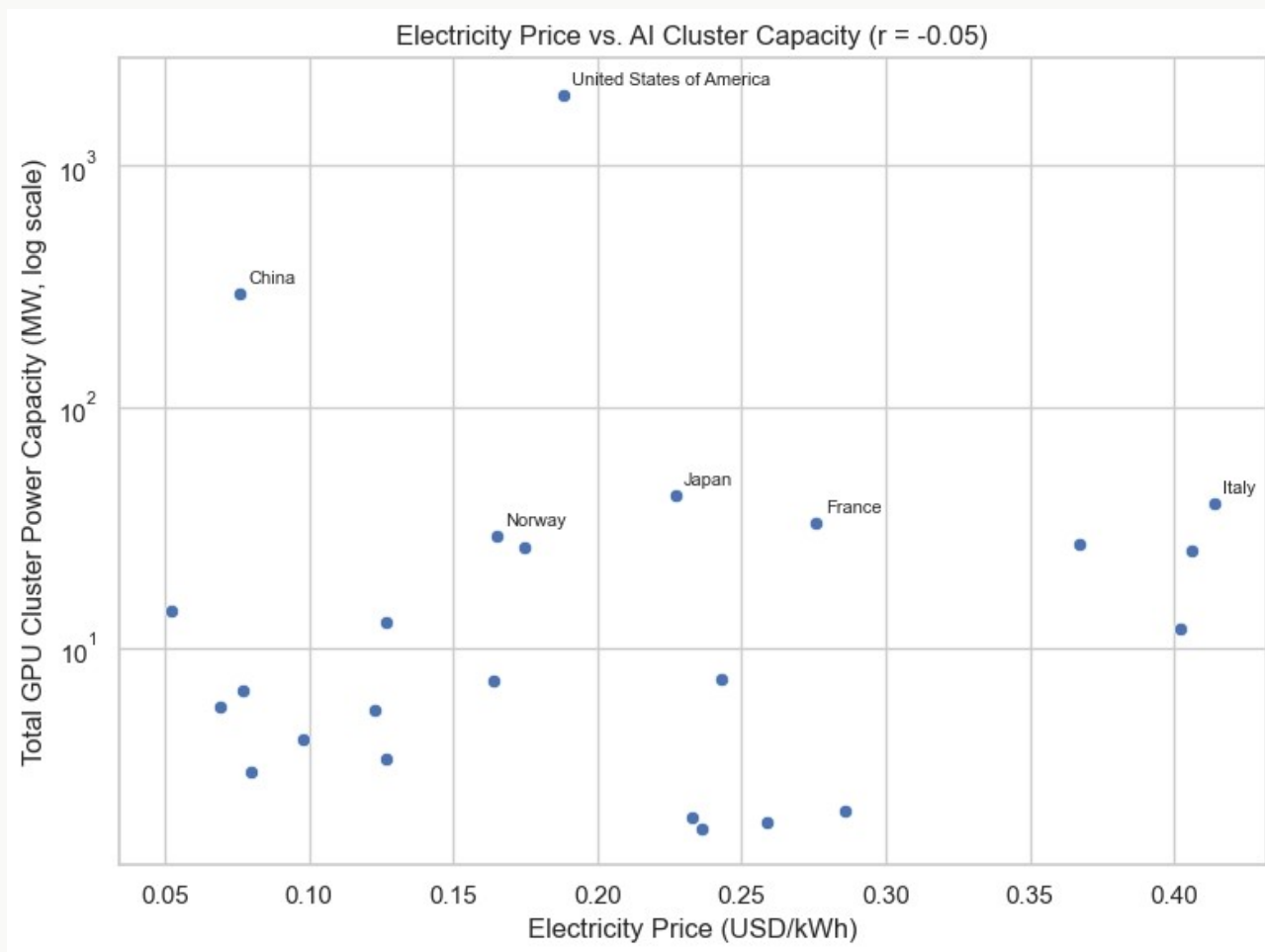
United States: 119

Japan: 25 · France: 14 · Germany: 11

But the US still leads on total power capacity (1,963 MW vs. 297 MW) — fewer, much larger US sites vs. many smaller Chinese ones.

This is `gpu_clusters_clean.csv` (482 sites, broad survey), which tells a very different story than the curated 74-site Epoch AI table used elsewhere (almost entirely US hyperscalers).

Cheap electricity does not predict AI capacity

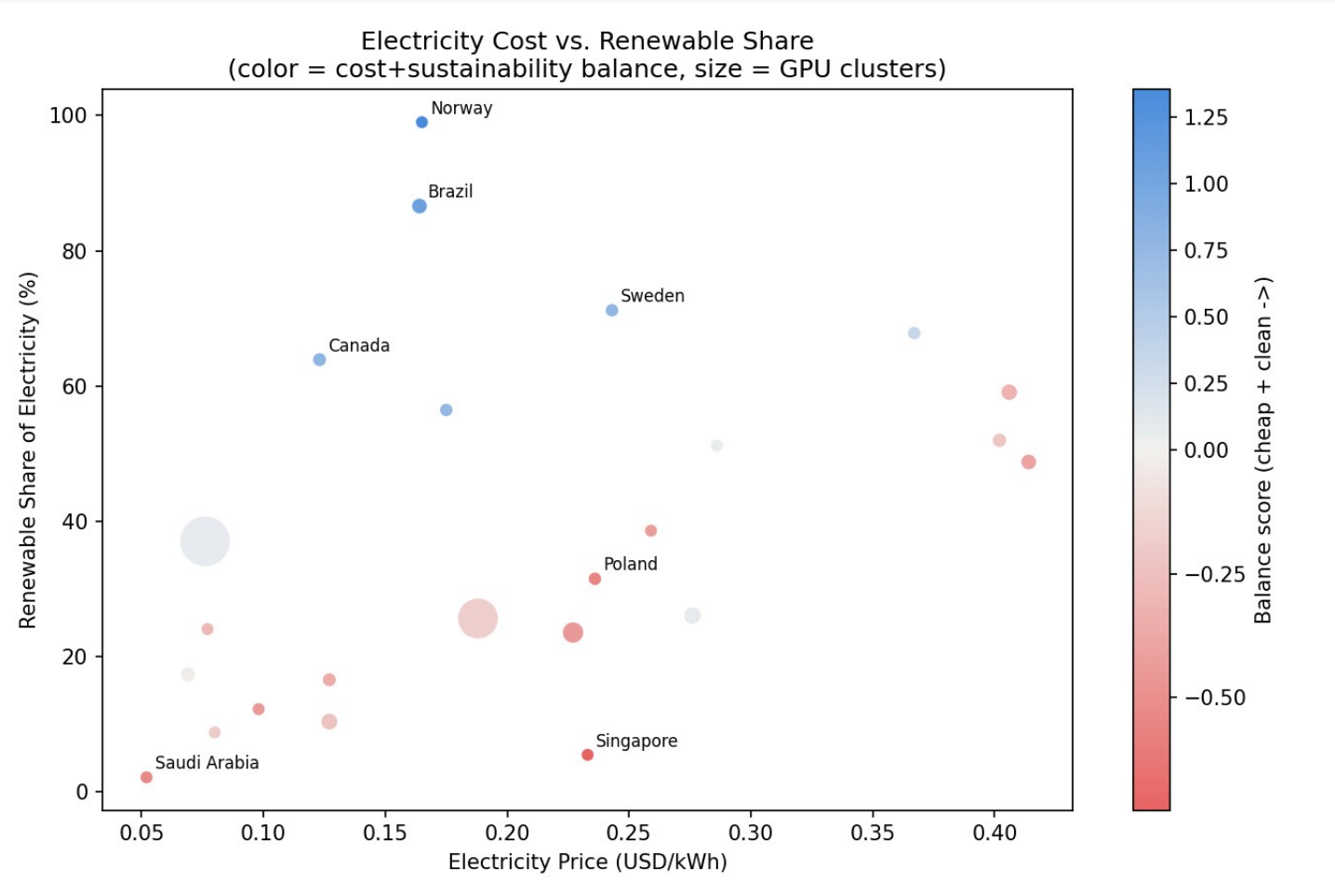


$r = -0.05$ — essentially zero correlation.

The US pays 0.188 USD/kWh, more than Russia (0.069), India (0.077), Taiwan (0.098), and Saudi Arabia (0.052) — all of which host far less AI capacity.

Takeaway: whatever drives where AI infrastructure concentrates, it isn't raw electricity price. Capital availability, existing cloud ecosystems, chip supply access, and policy plausibly matter far more.

Cost and sustainability don't reward the same countries

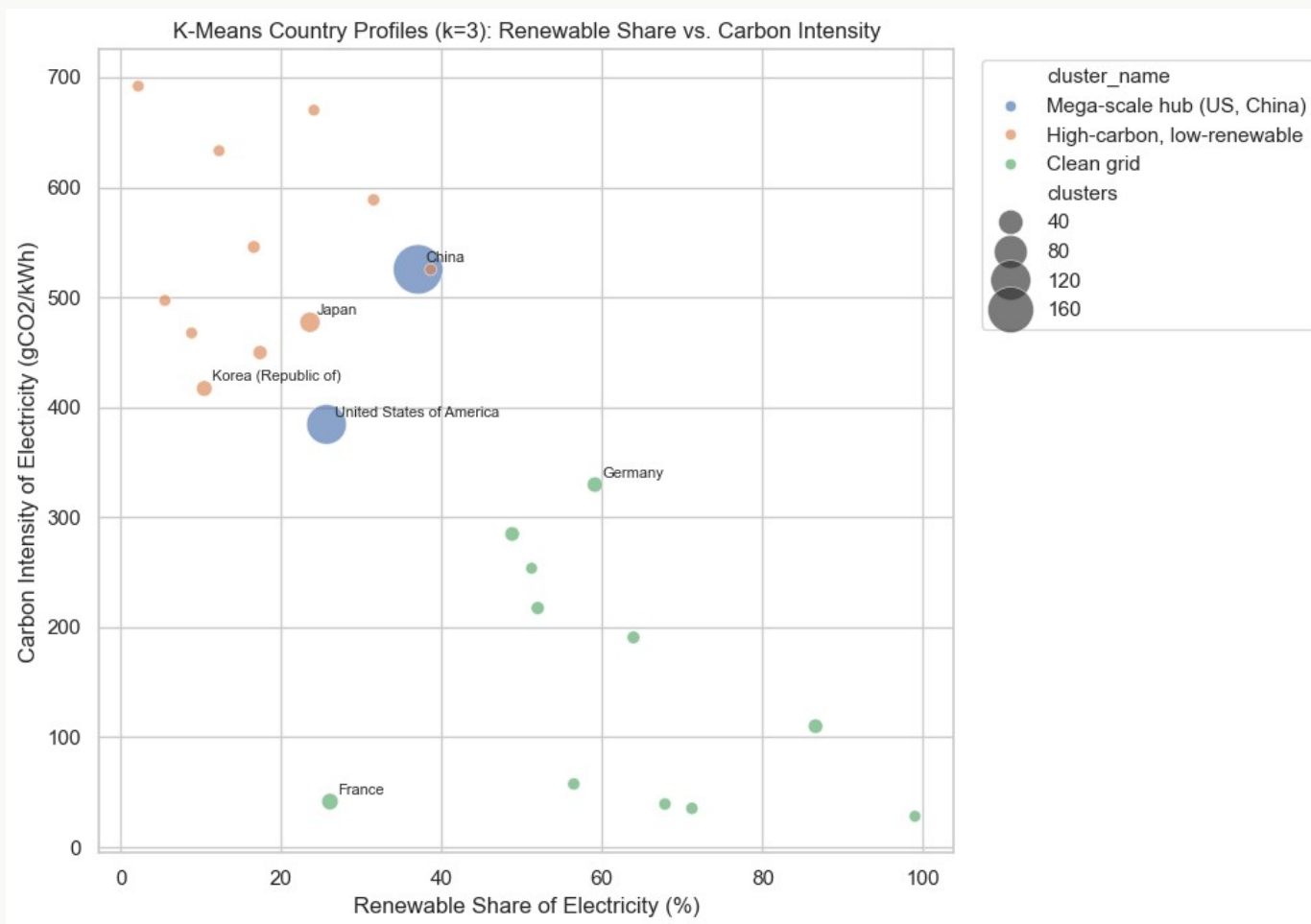


Norway and Brazil lead a composite score (cheap electricity + high renewable share + low carbon intensity).

China ranks notably higher here than on sustainability alone — it's genuinely cheap (0.076 USD/kWh) even though its grid isn't especially clean.

The US ranks below average on this composite despite hosting the most capacity by far — today's build-out doesn't track this balance.

Scale — not sustainability — is what separates countries

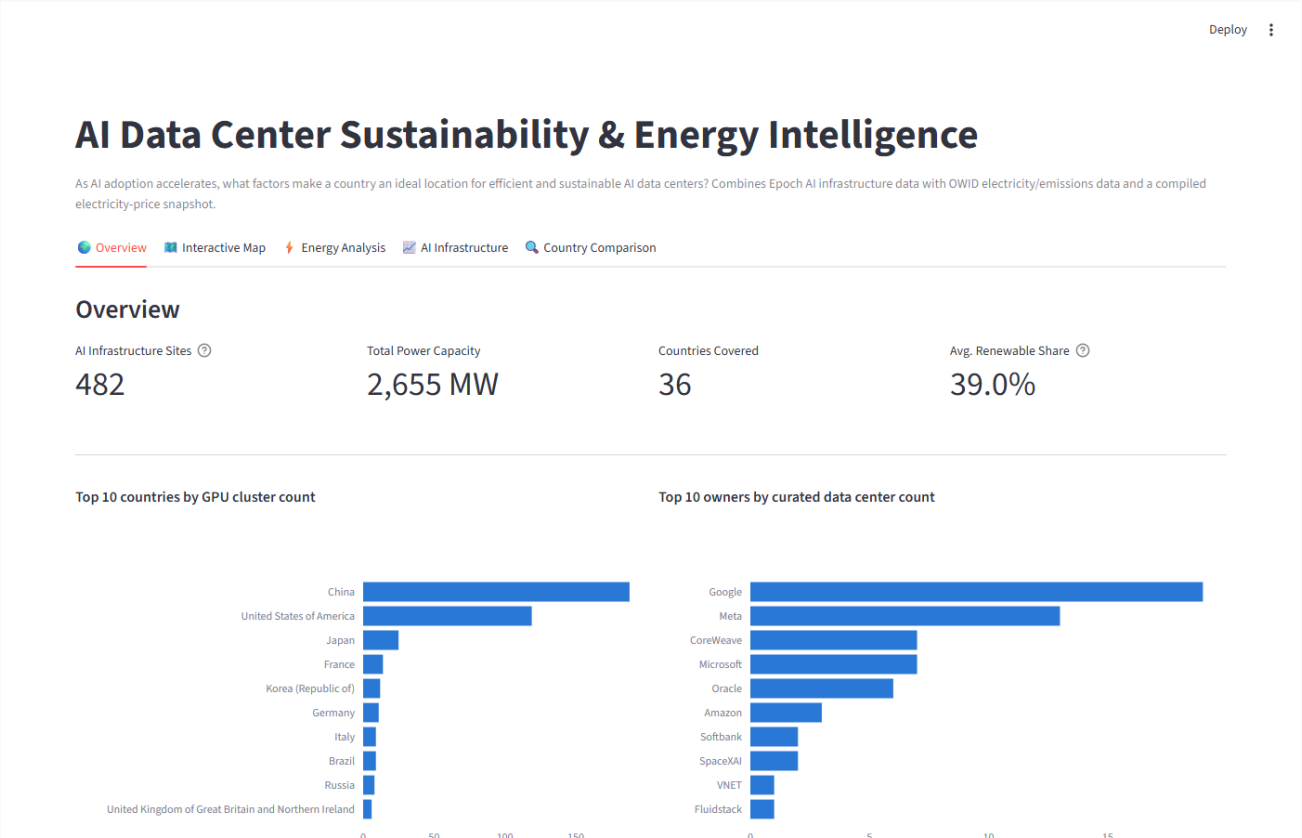


K-Means (k=3, silhouette-selected) finds:

- Mega-scale hub — just US & China, separated by raw scale alone
- Clean grid — 11 countries, avg 62% renewable
- High-carbon, low-renewable — 11 countries, avg 17% renewable

Today's two largest AI hosts aren't chosen for grid cleanliness — they're just where the scale already is.

Interactive dashboard



5 tabs: Overview · Interactive Map · Energy Analysis · AI Infrastructure · Country Comparison

What each tab does

 Overview

KPI cards + top-10 breakdowns by country and owner

 Interactive Map

True choropleth (log-scaled) + filterable site-level map

 Energy Analysis

Price, renewable share, and CO2-weighted emissions bubble chart

 AI Infrastructure

Capacity trend, provider treemap, cost+sustainability ranking

 Country Comparison

Pick 2+ countries → table + normalized radar chart

CONCLUSIONS

What actually predicts a good AI data center location

- Scale, not sustainability, explains where AI infrastructure sits today — the US and China form their own cluster purely on raw capacity.
- Electricity price shows no meaningful relationship with where capacity concentrates ($r = -0.05$).
- Renewable share alone is a misleading “clean grid” proxy — France's nuclear-heavy grid beats most higher-renewable countries on actual carbon intensity.
- If sustainability and cost were the deciding factors, Norway and Brazil — not the US or China — would be where new capacity goes.
- 24x spread in carbon intensity (Norway 28 vs. Saudi Arabia 692 gCO₂/kWh) among countries already hosting meaningful AI infrastructure — there's real room to choose better.

Known limitations

- No live electricity price feed — the price data is a manually compiled, point-in-time snapshot (2023-2026 average), not a live API.
- Two AI infrastructure tables tell different stories: the curated 74-site Epoch AI table is heavily US-weighted; the 482-site GPU cluster survey is broader but less curated.
- ~60 of 482 GPU cluster sites have no reported power capacity — they appear on the map but contribute 0 to capacity totals.
- CO2 emissions figures are national totals, not attributable to AI infrastructure specifically.
- Country-level analysis (24 countries with 3+ clusters) trades off granularity for statistical reliability — smaller markets are excluded from rankings.

Links

GitHub repository

https://github.com/many-25/AI_Data_Centers_Analytics

Live dashboard (Streamlit Community Cloud)

<< ADD URL AFTER DEPLOYING — see deployment steps >>

Analysis notebook

[notebooks/02_EDA.ipynb](#) (HTML/PDF export included in repo)

Thank you